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Introduction to RECOMMENDER SYSTEMS

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Lecture outline

Topics	Activities
• Define RSs • Introduce main vocabulary • Analytical framework • Applications of RS • **Non-personalised recommenders** • Example implementations	• Tasks – analyse and discuss in groups • Q&A

“Getting information off the internet is like taking a drink from a fire hydrant”

Mitchell Kapor

What are Recommender Systems (RS)?

- Belong to a class of **information filtering systems**
- Algorithms/tools/techniques that:
  - Take user's preferences as an input
  - Predict a rating that a user would give to an item
  - Output suggestions/recommendations about what a user may like
- Beyond algorithms, these are **systems**. That combine:
  - machine learning/data mining
  - business and marketing
  - HCI

Sources: Milano (2019), Schafer (1999), Jain (2015), Paraschakis (2016)



www.bigcommerce.com › blog › best-ecommerce-stores

40 Best Ecommerce Stores: Get Inspired With Designs We ...

What Makes An Awesome **Online Store**. Easy site navigation. Easy checkout, shipping options and payment process. Quality website design. High-quality photos and excellent copy for product descriptions. Product reviews help sales. Fast load times and a great mobile experience. Great customer support.

What Makes An Awesome... · Expert Opinions On What...

en.wikipedia.org › wiki › Online_shopping

Online shopping - Wikipedia

Online shopping is a form of electronic commerce which allows consumers to directly buy goods or services from a seller over the Internet using a web browser ...

hbr.org › 2011/12 › the-future-of-shopping

The Future of Shopping - Harvard Business Review

She buys one item from Danella **online** and then drives to the Danella **store** near her for the in-stock items she wants to try on. As Amy enters Danella, a sales ...

www.shopify.com › tour

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Everyday finds



Face masks →



Wall decor →



Gift Ideas →



Outdoor & garden →



Self-care →



Craft kits →

Popular right now



Monogrammed boys easter r...
SweetAndStitched
★★★★★ (2,992)
USD 38.50
Bestseller



Bamboo Lid Reusable Bubbl...
reusablebbtoup
★★★★★ (884)
USD 18.20
Bestseller



Metal Birds on a Branch - Lov...
WeAreMadAboutMetal
★★★★★ (252)
USD 30.63
Bestseller



Flower pot grow - 'Thank yo...
SelfmadeStudioDe
★★★★★ (324)
USD 11.34
Bestseller



Cornish Sea Breeze Reed Diff...
BeautifullyHC
★★★★★ (766)
USD 27.72
Bestseller

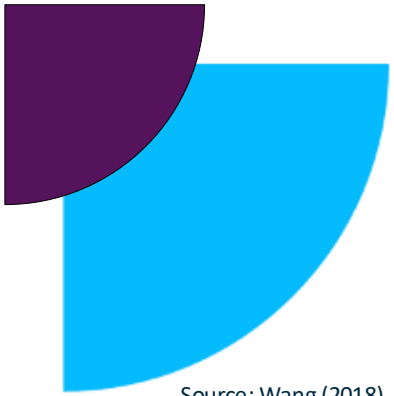


Smash Cake Outfit Boy Birth...
outeasme
★★★★★ (5,795)
USD 8.00
Bestseller

A little history

- Mid-1990's first RSs pioneered
  - personalized **navigation** system “Web Watcher”
  - 1995, Lieberman MIT
    - personalized **navigation** agent “Letizia”
  - 1997, AT&T Labs
    - **collaborative filtering** - “PHOAKS” (People Helping One Another Know Stuff) and “Referral Web”
  - 1999, Dresden University of Technology
    - personalized **e-commerce** system “TELLIM”
- 2001 - IBM
  - personalized features to the **e-commerce platform** - “Websphere”
- 2003, Google “AdWords”
- 2006 - Received attention after the Netflix Prize
  - contest that crowdsourced recommendation algorithms from research community
  - surpass Netflix’ own algorithm’s accuracy (RMSE) by 10%

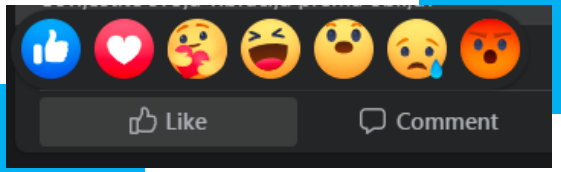
- **What are the current trends in RS?**



Source: Wang (2018)

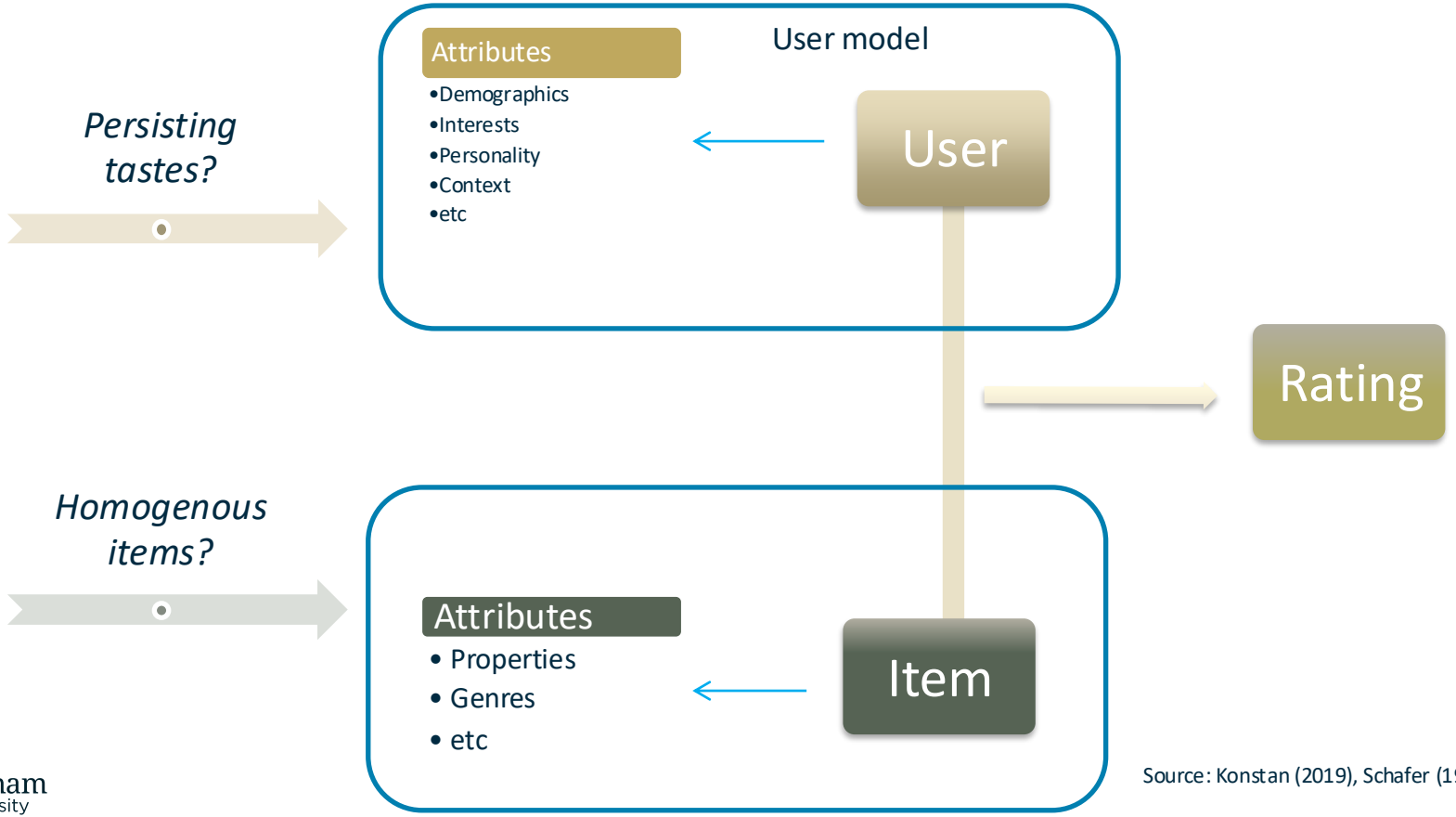
Threshold concepts

- **User model / profile**
  - Structured representation of user interests
- **Item's content**
  - Attributes/text describing an item
- **Rating**
  - Expression of preference
- **Prediction**
  - Estimate of preference or rating
- **Recommendation**
  - Suggestions of selected items



Source: Konstan (2019)

User-Item -> Rating



Source: Konstan (2019), Schafer (1999)

Predictions vs. recommendations

Prediction

- helps quantify
- a numerical value
  - e.g., scale from 1 to 5
- predicting liking (rating) of an item for the active user

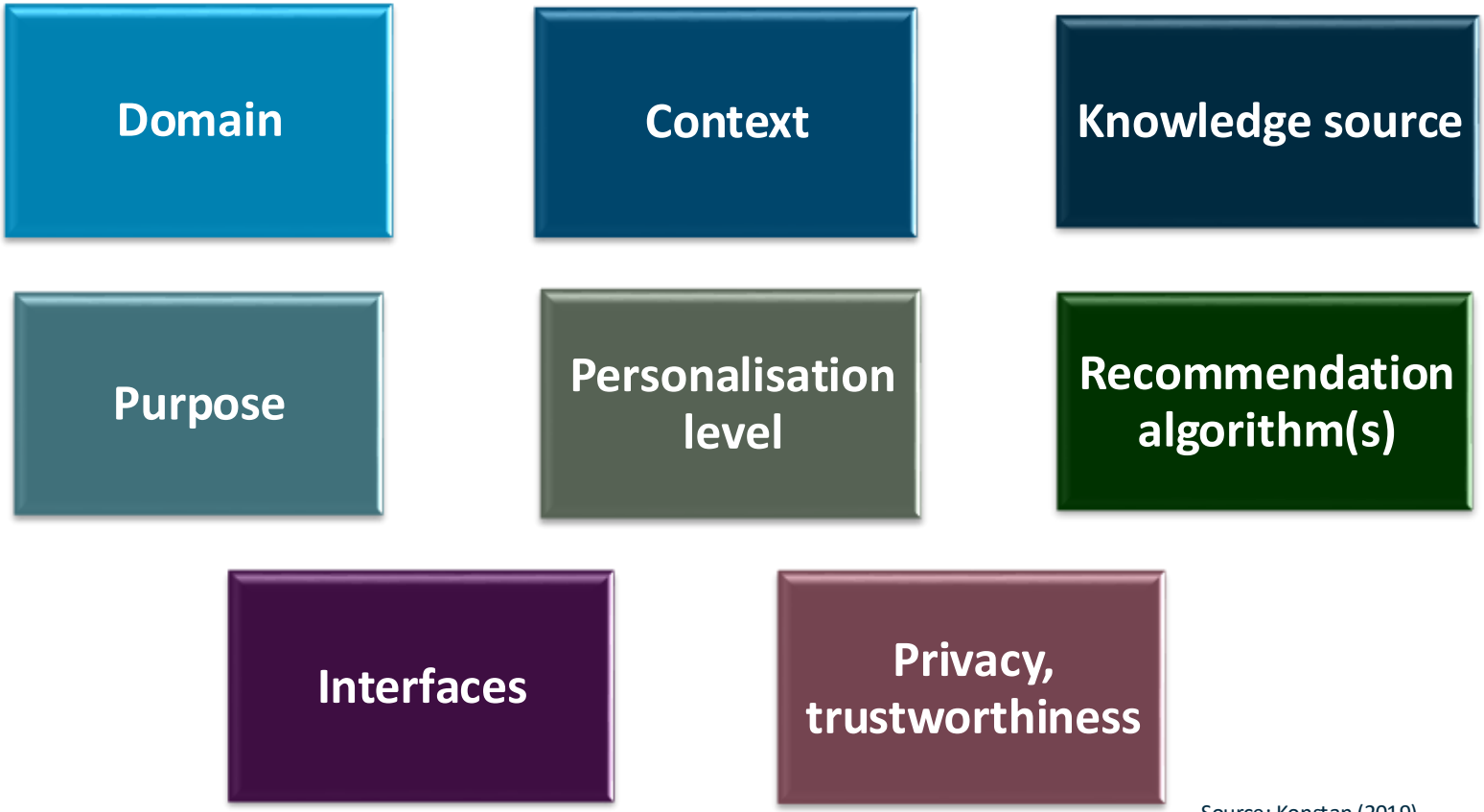
Recommendation

- selected items for the user from a larger set
- E.g., N items the active user would prefer the most
- Top N items or unordered

Source: Konstan (2019), Sarwar (2001)

Analytical framework

Analytical Framework Dimensions



Source: Konstan (2019)

Purpose / Objectives

Content recommendations

- **Sales**
  - Turn browsers into buyers - help customers find products they wish to purchase
  - Cross-sell - suggesting additional products for the customer to purchase
- **Information**

Education of users

- **Educate** about making the right decision, not steer us towards a particular decision (e.g. TripAdvisor)

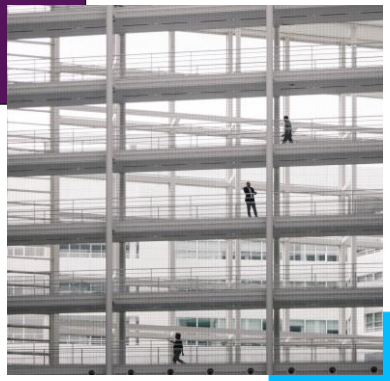
Community building

- **Loyalty** - improve loyalty by creating a value-added relationship between the site and the customer
- Join **community** to also give feedback (e.g. TripAdvisor)

Source: Konstan (2019), Schafer (1999)

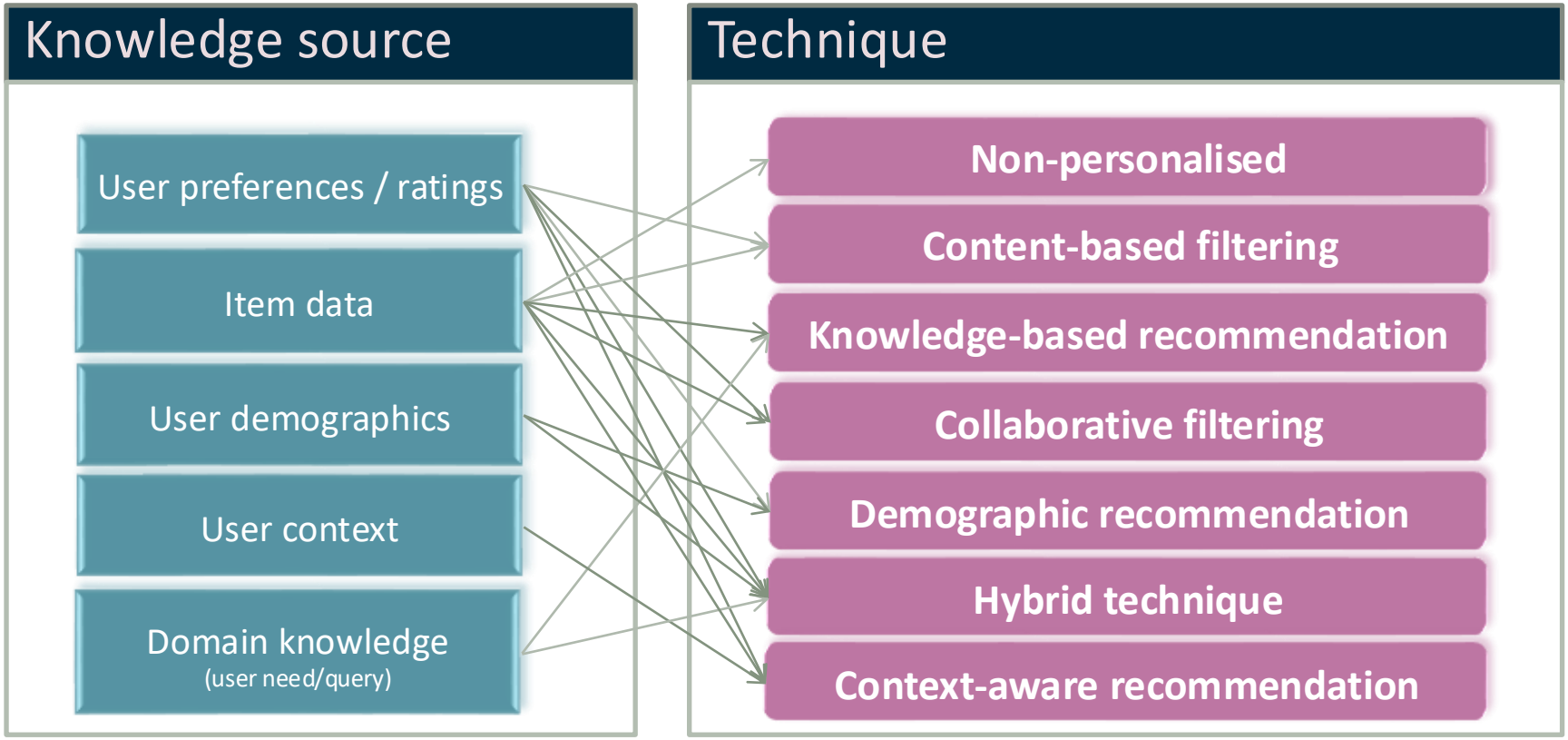
Levels of personalisation

- **Generic / non-personalised**
  - Same recommendations to all users
- **Demographic**
  - Matching a demographic group
- **Ephemeral**
  - Matching current activity
- **Persistent**
  - Matching long-term interests



Source: Konstan (2019)

Recommendation Techniques



Source: Konstan (2019), Burke (2007)

Interfaces: Input

Preference expression

Explicit

- Like / thumbs up (binary)
- Rating (numeric / star scale)
- Review / feedback / comment
- Vote
- Profile data: demographics, interest expression
- Etc.

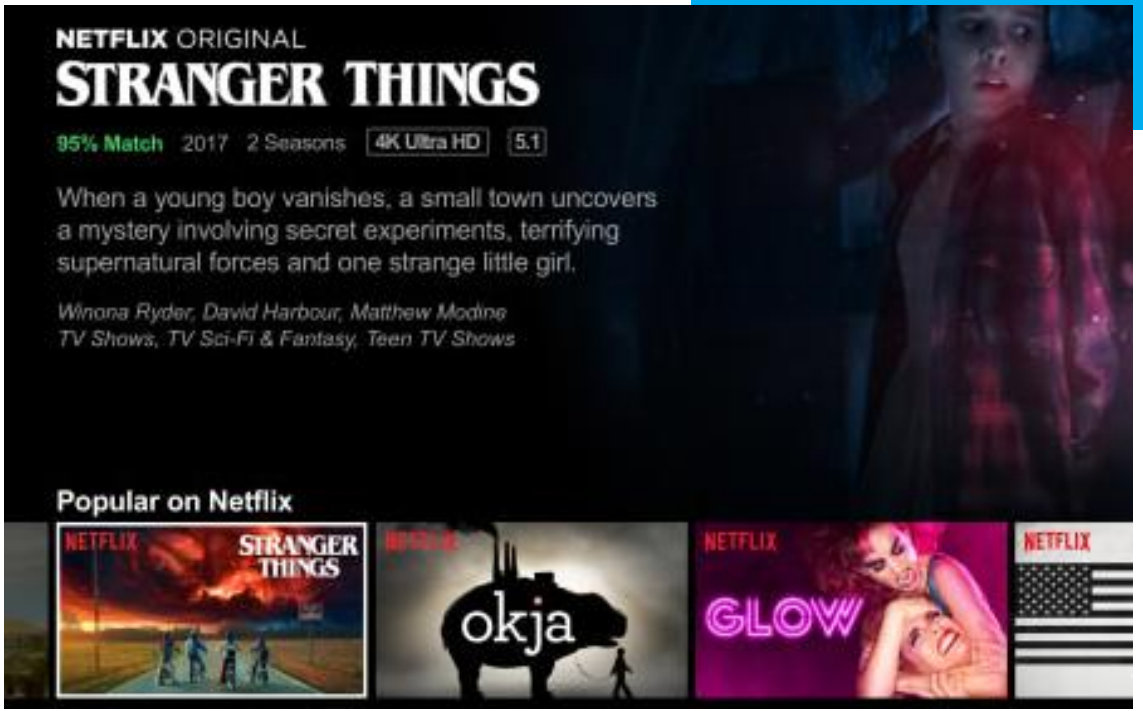
Implicit (inferred)

- View, click
- Order, purchase
- Following
- Saving, deleting, printing, bookmarking
- Etc.

Source: Konstan (2019), Lops (2010)

Interfaces: Output

- Predictions
  - e.g. TripAdvisor
- Recommendations
  - E.g. YouTube
- Filtering
  - e.g. email
- Explanations



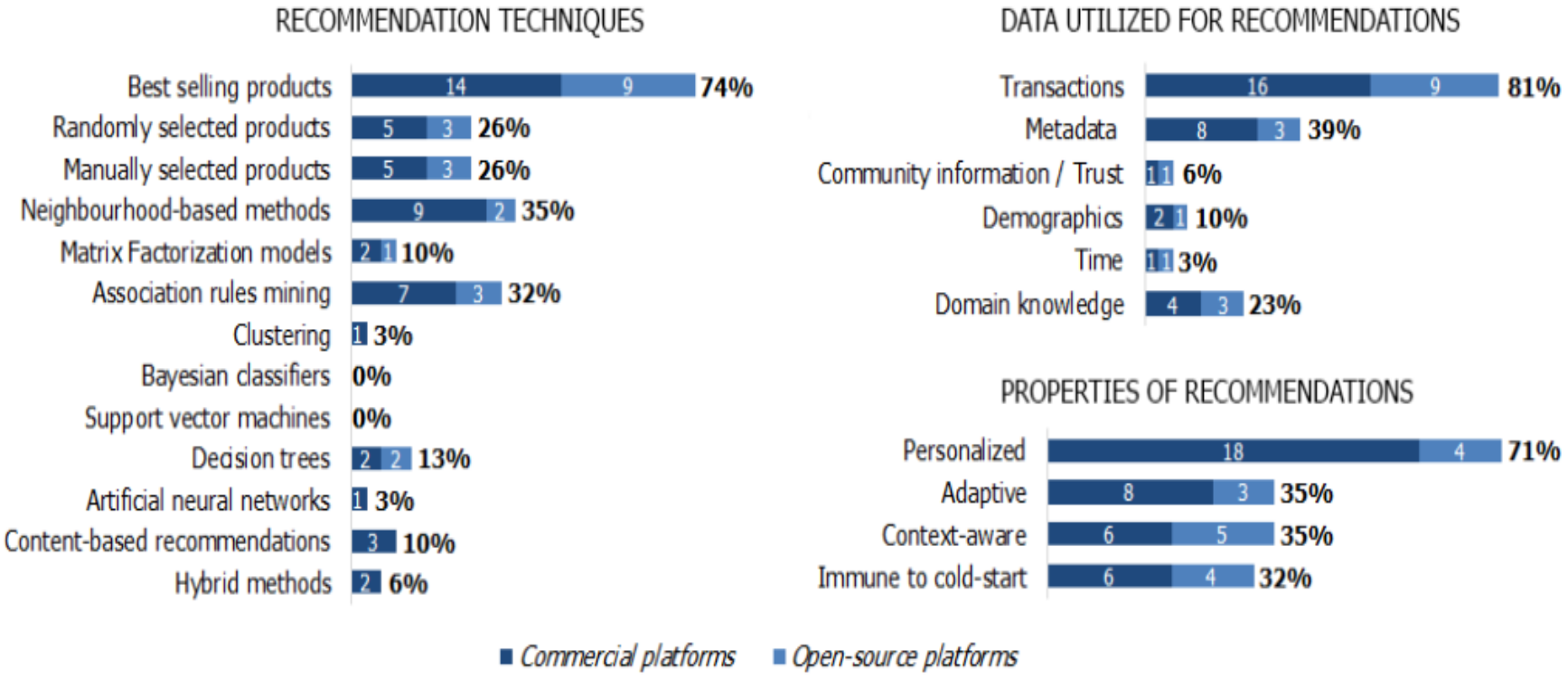
Source: Konstan (2019), Lops (2010)

Challenges
Privacy, Trustworthiness

- Privacy
  - Vast amount of personal information required
  - Deniability of preferences
- Bias (e.g., due to business rules)
- External manipulation (e.g., hacking)
- Explainability, transparency
- Cold-start: new items, new users
- Changing user preferences
- Unpredictable items

- Over-specialisation
  - Lack of diverse recommendations
- Sparsity
  - Lack of knowledge
- Scalability
  - Growing number or users and/or items – more resources needed
- Seasonality effects
  - Need for time-based dataset splits
  - (e.g. fashion, travel, etc.)

- Findings for **e-commerce recommenders**
  - Paraschakis, D. (2016, September). Recommender systems from an industrial and ethical perspective. In *Proceedings of the 10th ACM Conference on Recommender Systems* (pp. 463-466). ACM.



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Non-personalised RS

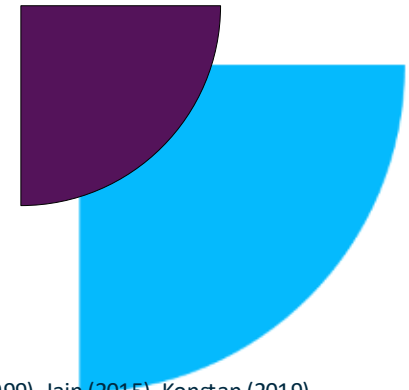
What?

- Generic recommendations are:
  - Based on what other users have said about items on average
  - Independent of the user
  - All users receive the same recommendations

Source: Schafer (1999), Jain (2015), Konstan (2019)

Examples of generic recommendations

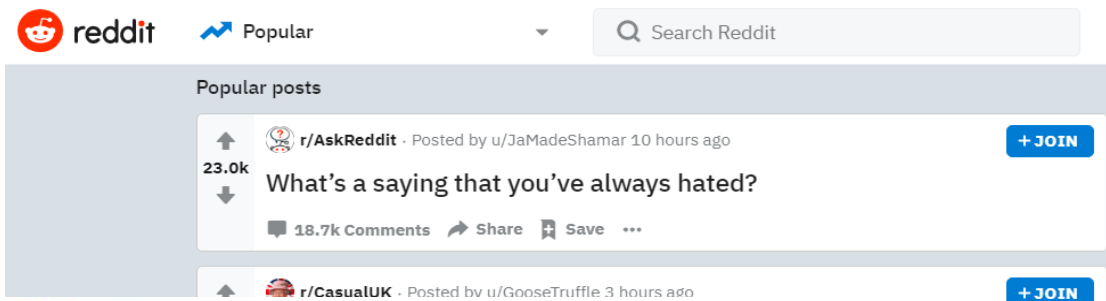
- **Product associations**: Customers who bought X also like…
  - not personalised to an individual, but connected to the context of an item currently viewed or ordered
- Most **popular** items
  - with respect to the number of ratings within a group / population
- **Similar** items
- **Trending**
- **Latest** items / new releases
- Stereotyped
  - popular within a target group (e.g. young females from the UK)

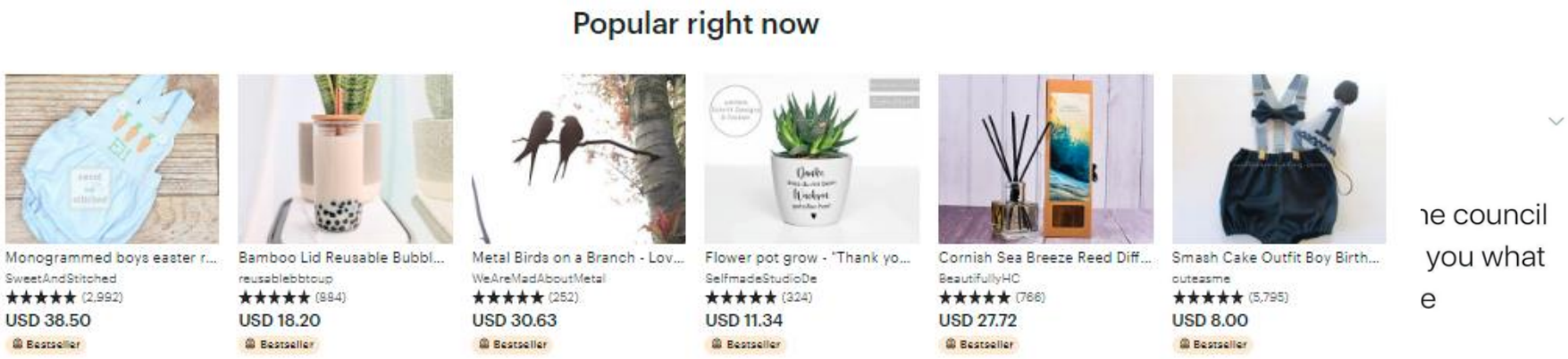


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Source: Schafer (1999), Jain (2015), Konstan (2019)

Example applications







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Key topics to take away

- Main concepts
  - recommendation vs. prediction, user-item -> rating, user model/profile
- RSs definition
  - tools that take a user's preferences as an input, predict a rating that a user would give to an item, and output recommendations about what a user may like
- Dimensions of analytical framework
  - domain, purpose, context, knowledge source, personalisation level, recommendation algorithm, privacy, interface
- Generic RS
  - All users receive the same recommendations, based on averaged other users' preferences
  - examples: trending, most popular, new releases, etc.

References and reading material

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Task

- In groups of 2-3
  - Choose one commercial RS; it should be from a domain you would like to develop your RS for
  - Use the analytical framework to determine its:
    - Domain
    - Purpose
    - Recommendation context
    - Knowledge source
    - Personalisation level
    - User interface
      - input (explicit/implicit data)
      - output (what is presented to users and how)
    - Recommendation techniques
      - if not straightforward, do a little online research
- Post your solutions **individually** on this week's discussion board.

Task

- In groups of 2-3
- Choose 1 commercial RS from a **domain you would like to develop your RS** for
  - E.g., Etsy, Netflix, Amazon, Spotify, Zalando, YouTube, LinkedIn, Blinkist, etc.
- Identify and describe the following:
  - User
  - Item
  - Rating
- Does the RS display predictions? How?
- What are recommendations? How are they presented?
- Post your solutions **individually** on this week's discussion board.